dimer rows. This is most probably due to the overlapping of the antibonding orbitals of C–C dimers[8]. It follows that the surface electronic image states[17] can be considered to be delocalized one-dimensional (1D) electronic states[18] along the dimer rows.

Finally, it is puzzling that we were able to transmit relatively high electron currents (1 nA) through a thick (0.2 mm) insulating diamond single crystal without any noticeable charging effect. This may be explained by an outstanding property of diamond single crystals; very long electron diffusion lengths of 150–250 μm have been measured[19]. Thus, diamond is insulating because of the lack of charge carriers, and yet electrons injected via the STM tip into the conduction band can be transported over very long distances without being trapped by defects.

We have used resonance electron tunnelling to image insulating diamond surfaces and to probe their electronic properties at the atomic scale using the STM. This approach could be used with other insulating materials provided they also have similar properties, that is, a large electronic band gap and long diffusion length of electrons in the conduction band. With these new capabilities for imaging insulating diamond surfaces at the atomic scale, surprising electron transport properties have been inferred, such as in 1D transport through the surface image states of dimer rows or in the injection of electrons in the conduction band. These results demonstrate that insulators like diamond could be advantageously used, beyond their well known passive dielectric properties, for active atomic-scale electronic devices. □

Received 21 March; accepted 13 August 2001.

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Acknowledgements

We thank P. Bergonzo and A. Hoffman for providing facilities to hydrogenate the diamond samples. This work is supported by the AFIRST programme, the European TMR network "Manipulation of individual atoms and molecules", and the European IST-FET "Bottom-up-Nanomachines", (BUN) programme.

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Selective assembly on a surface of supramolecular aggregates with controlled size and shape

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The realization of molecule-based miniature devices with advanced functions requires the development of new and efficient approaches for combining molecular building blocks into desired functional structures, ideally with these structures supported on suitable substrates[1–4]. Supramolecular aggregation occurs spontaneously and can lead to controlled structures if selective and directional non-covalent interactions are exploited. But such selective supramolecular assembly has yielded almost exclusively crystals or dissolved structures[5]; the self-assembly of absorbed molecules into larger structures[6–8], in contrast, has not yet been directed by controlling selective intermolecular interactions. Here we report the formation of surface-supported supramolecular structures whose size and aggregation pattern are rationally controlled by tuning the non-covalent interactions between individual absorbed molecules. Using low-temperature scanning tunnelling microscopy, we show that substituted porphyrin molecules adsorbed on a gold surface form monomers, trimers, tetramers or extended wire-like structures. We find that each structure corresponds in a predictable fashion to the geometric and chemical nature of the porphyrin substituents that mediate the interactions between individual adsorbed molecules. Our findings suggest that careful placement of functional groups that are able to participate in directed non-covalent interactions will allow the rational design and construction of a wide range of supramolecular architectures absorbed to surfaces.

A large number of different selective and directional intermolecular connections has been developed in supramolecular chemistry[5]. Here we use a cyanophenyl substituent to control molecular aggregation, because it has a simple and symmetric structure as well as an asymmetric charge distribution at the cyano group that should introduce dipole–dipole interactions

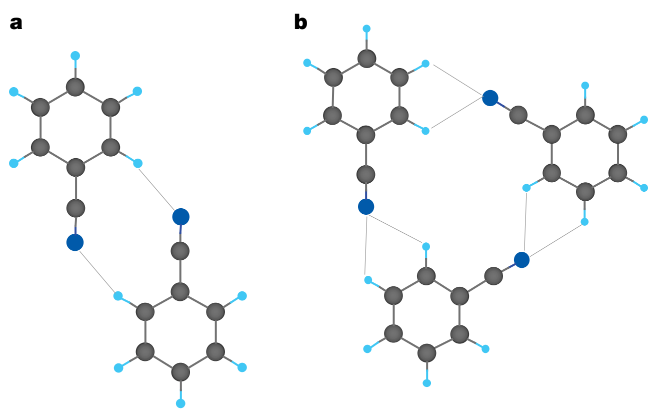

Figure 1 Calculated molecular aggregations. **a**, Cyanobenzene dimer; **b**, trimer. Colours correspond to the elements: H, light blue; C, black; N, dark blue.

between neighbouring cyanophenyl substituents. An illustration of this interaction is given in Fig. 1, which shows the optimized arrangement of a cyanobenzene dimer and trimer obtained in *ab initio* molecular orbital calculations at the MP2/6-31G* level. In the dimer, the cyano groups have an antiparallel configuration, whereas the trimer structure results from a cyclic arrangement of the cyano groups. The length of the CH…NC contacts is about 2.39 Å and 2.65 Å for the dimer and the trimer, respectively, and thus is shorter than the van der Waals distance of about 2.7 Å. The interaction energies are estimated to be $-7.12\,\mathrm{kcal\,mol^{-1}}$ and $-12.40\,\mathrm{kcal\,mol^{-1}}$ for the dimer and trimer structures, respectively, and thus comparable to the energy of hydrogen-bonding interactions. The relative orientation of molecules in these aggregates is therefore likely to be influenced by long-range dipole–dipole interaction, with hydrogen-bonding interactions further stabilizing the structure.

The characteristic intermolecular interaction between cyanophenyl groups is exploited to influence the aggregation of adsorbed porphyrin molecules. The porphyrin synthesized in this study is based on 5,10,15,20-tetrakis-(3,5-di-tertiarybutylphenyl) porphyrin (H_2-TBPP), which has a free-base porphyrin core and four di-tertiarybutylphenyl (tBP) substituents (Fig. 2a)[9–13]. To ensure selective molecular aggregation, we used a gold (111) as the substrate surface because of its inertness and its properties of reconstruction (see below). The surface was prepared by repeated cycles of Ar^+ sputtering and annealing in an ultrahigh-vacuum (UHV) chamber, followed by deposition of porphyrin molecules at room temperature by sublimation from a Knudsen cell in UHV. The resultant molecular arrangements were characterized by low-temperature scanning tunnelling microscopy (STM) carried out at 63 K (ref. 14).

Figure 3a shows an STM image of the Au(111) surface after a small amount of H_2-TBPP was deposited, with isolated single molecules located at elbows that appear in the surface-reconstructed pattern of the Au substrate. The reconstruction of the Au(111) surface is known to include 'herringbone' patterns[15], and a preferential nucleation of adsorbates at the elbows of this pattern has been reported for various systems[8,16]. A high-resolution STM image (see Fig. 3e) shows that the individual H_2-TBPP molecules are composed of four paired lobes surrounding two oblong protrusions. On the basis of the molecular dimensions, each lobe was assigned as one of the tertiarybutyl substituents, and the oblong protrusions as the central porphyrin ring. The appearance of the paired lobes suggests that the molecular plane of the phenyl rings is oriented to be closely aligned with the porphyrin mean plane, whereas about 90° rotations of the phenyl rings are expected for the ideal conformation of H_2-TBPP (refs 9–11). We estimate the dihedral angle between the porphyrin and phenyl rings to be about 20°, which results from the rotational flexibility of the porphyrin–phenyl bonds and from the optimum adsorption of the tertiarybutyl groups onto the Au(111) surface. Upon further deposition, the H_2-TBPP molecules exhibit a close-packed arrangement (Fig. 3e and i), which results from van der Waals interactions between the tBP substituents. (Further details of the adsorption of H_2-TBPP molecules on the Au(111) surface are given in ref. 17.)

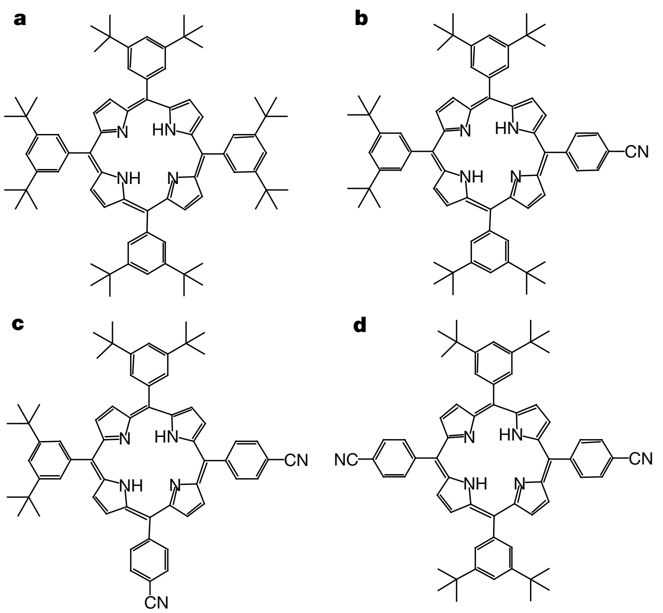

Figure 2 Structural formula of the porphyrins. **a**, H_2-TBPP, which is composed of a free-base porphyrin and four tBP substituents. In the ideal conformation, the phenyl rings are rotated by about 90° with respect to the porphyrin blane. **b**, CTBPP, where a tBP substituent of H_2-TBPP is replaced with a cyanophenyl substituent. **c**, *cis*-BCTBPP, where two cyanophenyl groups were substituted at the *cis* position. **d**, *trans*-BCTBPP, where two cyanophenyl groups were substituted at the *trans* position.

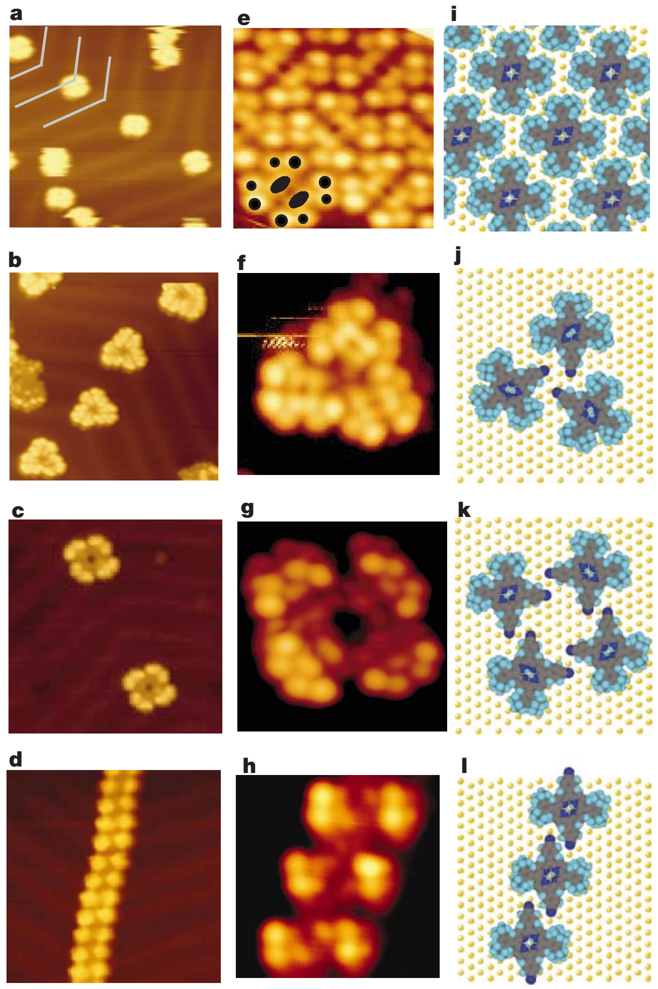

Figure 3 Scanning tunnelling microscope images at 63 K of supramolecular aggregations induced by the cyano groups. At small coverage, individual molecules or clusters are preferentially positioned at the elbows of the herringbone patterns of the Au(111) surface (imaged area, 20 × 20 nm): **a**, H_2-TBPP; **b**, CTBPP; **c**, *cis*-BCTBPP; **d**, *trans*-BCTBPP. High-resolution STM image (imaged area, 5.3 × 5.3 nm) and the molecular model: **e**, **i**, H_2-TBPP island; **f**, **j**, CTBPP trimer; **g**, **k**, *cis*-BCTBPP tetramer; **h**, **l**, *trans*-BCTBPP wire. As indicated by black dots in **e**, the STM image of the single H_2-TBPP molecule is composed of four paired lobes surrounding two oblong protrusions, corresponding to the tertiarybutyl substituents and the central porphyrin.

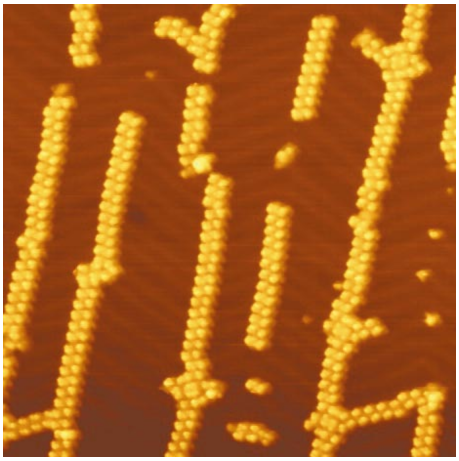

Figure 4 STM image at 63 K of *trans*-BCTBPP wires formed on the Au(111) surface (imaged area, 70 × 70 nm). The supramolecular wires extended across the elbows of the herringbone patterns, and the observed maximum length is above 100 nm, which depends on the terrace width.

To introduce the characteristic interaction of the cyano substituents, we have synthesized (cyanophenyl)-tris(di-tertiarybutylphenyl) porphyrin (CTBPP), where a tBP group of H_2-TBPP was replaced with a cyanophenyl substituent (Fig. 2b). At low coverage, most CTBPP molecules assemble into triangular clusters on the Au(111) surface. As shown in Fig. 3b, identical clusters are located separately at the elbows of the herringbone patterns. From the molecular structure of CTBPP, three paired lobes are expected as a single molecule in the STM image, because a paired lobe corresponds to one di-tertiarybutyl substituent. The high-resolution STM image of Fig. 3f shows that the triangular cluster is a CTBPP trimer. The cyanophenyl substituents are assembled into a cyclic configuration in the trimer structure of Fig. 3j, in agreement with the cyanobenzene trimer aggregation of Fig. 1b. Compared with the characteristic aggregation of CTBPP, we have not observed such supramolecular clusters for (phenyl)-tris(di-tertiarybutylphenyl) porphyrins, where the cyano substituent was synthetically removed from the CTBPP molecule. In this case, a random arrangement of the molecules was formed on the surface, confirming that the supramolecular aggregation is dominated by the cyano groups.

To probe the supramolecular aggregation in our system further, we substituted one more cyano group to give bis(cyanophenyl)-bis(di-tertiarybutylphenyl) porphyrin (BCTBPP), which forms two types of isomers (*cis* and *trans*) with respect to the configuration of two cyanophenyl substituents (Fig. 2c and d). Figure 3c shows that the *cis*-BCTBPP molecules are aggregated into a supramolecular tetramer. In this structure, the antiparallel intermolecular connections of all the cyanophenyl substituents lead to a macrocyclic arrangement of the porphyrin molecules, forming a molecular ring (see Fig. 3g and k).

In contrast to the macrocyclic clusters of CTBPP and *cis*-BCTBPP, sequential aggregation was achieved for the *trans*-BCTBPP molecules, where two cyanophenyl groups were substituted at the trans positions. Figure 3d and h shows the STM images of the Au(111) surface after submonolayer deposition of the *trans*-BCTBPP molecules. In this structure, the antiparallel configuration between the cyanophenyl substituents results in a linear arrangement of the *trans*-BCTBPP molecules, forming the supramolecular wires. As shown in the large-scan STM image of Fig. 4, most of the individual wires extended across the elbows of the herringbone patterns, because the molecules were initially nucleated at the elbows. Furthermore, it is remarkable that the maximum length of the straight wires is above 100 nm, although some branches are also formed due to the three-fold symmetry of Au(111).

These selective aggregations of the porphyrin molecules result from the antiparallel or cyclic configurations of the cyano substituents relative to each other. From the molecular models and the high-resolution STM images, we estimated the length of the CH…NC contact between cyanophenyl substituents to be about 2.6 Å and 2.5 Å for the antiparallel and cyclic configurations, respectively. These values are almost identical to those for the calculated cyanobenzene aggregates, suggesting hydrogen-bonding interactions. Although the strength of the non-covalent intermolecular interactions in our supramolecular aggregates is thus significantly weaker than encountered in covalent bonds, electronic coupling across hydrogen-bonded interfaces can nevertheless allow electron or energy transfer[18], as has been observed in biological systems[19,20]. In principle, systems similar to those described here should therefore be capable of exhibiting useful electronic and optoelectrical functions. Furthermore, the central porphyrins and cyanophenyl substituents within the clusters or wires should be decoupled from both the substrate surface and surrounding clusters in our present systems, because the bulky tBP insulators attached to the surface prevent any direct interaction. This indicates that unusual chemical and physical properties associated with the size and shape of the aggregate cluster should remain, even after deposition on the surface. □

Received 30 April; accepted 1 August 2001.

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Acknowledgements

T.Y. would like to thank K. Kimura, Y. Takaguchi and Y. Wakayana for useful discussions.

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